Varun Vikash Karthikeyan

Interface design for predictive maintenance in trucks

Research paper based on project:

(under review) **Ecological Interface Design and Emergent Users: Designing for Small-Scale Trucking Ecology in India**

Human Factors and Ergonomics in Manufacturing & Service Industries. Wiley

*Vivek Kant, **Varun Vikash Karthikeyan**, Nishant Sharma*

App walkthrough video
https://drive.google.com/file/d/1-THUZbE-1BsiOnRmjzv0K6BlpBfa7NPz/view?usp=sharing

Project Report (thesis)
https://drive.google.com/file/d/1zgNLbU3GkCHnMzChLW5L18OosSRlYIqK/view?usp=sharing

Context

This was my bachelor graduation project. It was an individual project done under the supervision of a guide and a co-guide.

Scope
About 75% of all trucks in the country are owned and operated by small-scale owners(owning less than 5 trucks). The trucks in general are in poor condition. Maintenance is usually carried out at local workshops where all parts except oils and filters are replaced only when they are completely worn out, owing to lower costs in the short term. But poor maintenance practices along with posing safety hazards, damages the health of the vehicle in the long run leading it to return lower mileage and cause more pollution.

The aim of this project is to take the concept of predictive maintenance (using sensors to continuously monitor the remaining life of parts so that they can be just before they fail) to the context of small-scale truck owners.

Popularity of smartphones
With smartphones being available at affordable prices, they are becoming increasingly popular even among people who have not owned any kind of computation device earlier. This combined with the availability of cellular data at cheap prices, it has become possible to provide digital services to people with whom it was not possible before the age of the smartphone.

Working of the system
The small-scale trucking ecology involves a brokerage system for functioning. The broker system, locally referred to as the 'mandi' system involves the owner having their trucks in a ready-to-travel condition parked at the broker's office and giving the broker one destination/route preference. When the broker gets an assignment to that particular destination/route, the trucks parked at the office are sent on the designated trip based on a first-come-first-served basis. The choice of where owners would want their trucks to be sent to is made largely based on personal preference with no solid rationale behind them. It can be based on places their trucks have been to earlier, places the trucks of people they know have been to, as well as others. Apart from the case of a total breakdown, any kind of maintenance activity is not undertaken during a trip as owners prefer to carry out maintenance only at their trusted local workshops back home. Hence, it is important for maintenance required for the truck to successfully make a trip to the preferred destination to be completed before the truck is sent to the broker's office.

Initial Brief
Design a smartphone interface to enable small-scale truck owners(<5 trucks) derive the benefits of predictive maintenance. . It is assumed that it is possible to have information from the sensors that are added at later stages to also be detected and be displayed on the interface.

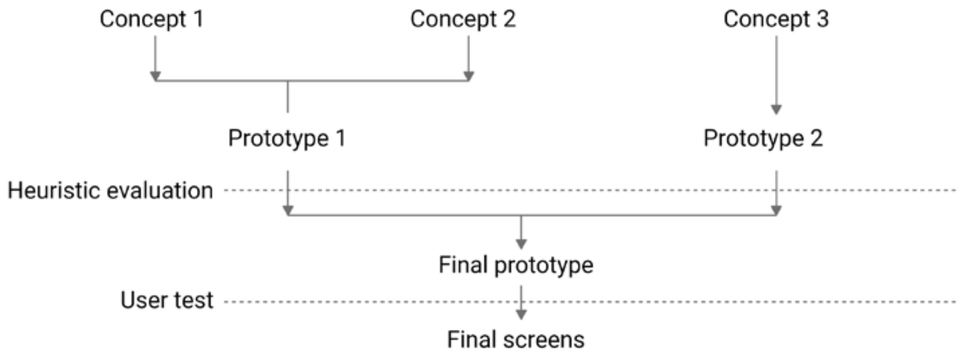
Process

User study
An extensive user study was conducted in Feb-Mar, 2020 in and around the areas of Salem, Sankagiri, Tiruchengode, and Thungavi in Tamil Nadu. A total of 21 people directly related to the small-scale trucking sector(<5 trucks) were interviewed (5 truck owners, 7 truck drivers, 8 mechanics, 1 spare parts dealer; one of the truck owners was also a broker) by means of face-to-face interviews that lasted for about 30-45 min each.

User study

Key Insights
1. With the absence of the helper, there are certain things like engine oil level, radiator water level, tyre pressure that need to be constantly monitored by the driver.
2. Once the truck returns from a trip, the owner gives the broker a preference of where to go next. There is no fixed logic/method to this preference.
3. The owner is fully dependent on the driver to be informed of faults for maintenance to be carried out.
4. The availability of ready-at-hand cash or spare parts does affect maintenance schedules.
5. Not all owners are familiar with inputting text onto the smartphone.

Re-defined brief
Design a smartphone interface with which the small-scale truck owner(<5 trucks) can:
1. See if running conditions are ensured by the driver
2. View health of parts and plan for maintenance
3. Decide on next trip based on condition of truck



Conception Process

Designing the screens
The design process was iterative in nature and consisted of creation of three design concepts that differed on the creative design basis and not on the analytical basis; i.e., they differed on the creative ways in which the information was presentedConcept 1 was exploratory and helped bring out the initial ideas. In Concept 2, the emphasis was to make things as simple as possible and provide only the information that is absolutely necessary. In contrast to these two, Concept 3 took a more experimental approach in the form of a landscape orientation (1 & 2 were portrait) with the aim to give the user rich information for decision making. Later, Concepts 1 and 2 were combined to make Prototype 1(portrait) while Concept 3 was developed into Prototype 2.

Evaluating the Design
An initial heuristic evaluation was carried out on both the wireframe prototypes with 5 final semester design students. This step ensured that usability heuristics were adhered to during the design. A detailed high-fidelity prototype interface was developed based on these inputs. Next, usability evaluation of this detailed prototype was carried out with 8 small-scale truck owners in and around the areas of Mallasamudram and Tiruchengode in Tamil Nadu, India. Out of these people, 6 of them had a personal smartphone and the other 2 had a smartphone in the house, but it was not their primary phone.

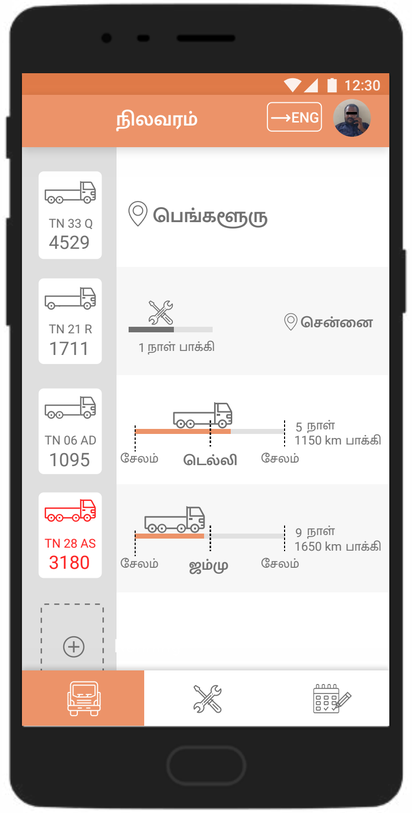
The usability test was made up of 3 parts:
First, participants were asked to identify the icons used to represent each of the parts from a sheet of paper. This was necessary because all the parts were represented primarily with visuals/icons.
Second, participants were asked to perform 10 predefined tasks on the interface. The tasks were chosen in such a way that they covered the most common use-cases of the app.
Third, participants were asked to select 3 places on the 'map-based location selector' - a novel mode of interaction developed specifically for this context (see section 5.4.4). The test results indicated that users found it easier to identify icons of parts/parameters whose representation was close to how they actually look like in the truck as opposed to more commonly used/abstract versions of representation. One major problem in the interface was that some users were not able to relate to the exclamation (!) as an indicator of a fault/warning. Upon being asked, they revealed that on the dashboard of the truck, exclamation (!) stood for 'handbrake engaged' and faults were communicated through blinking icons. Hence, they suggested that blinking icons would be a better representation of faults/errors. Apart from this, the tests highlighted a few areas where the grouping of elements had to be improved. These insights were incorporated to formulate the final interfaces (Figure 6).

Outcome and Key Rationales

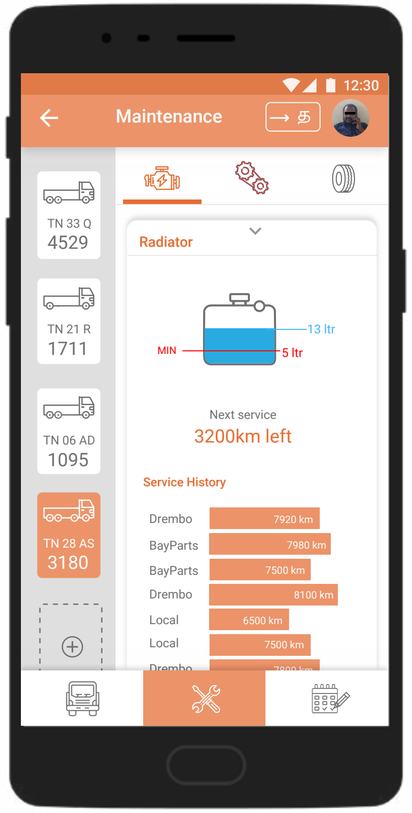
Navigation of the interface
The final design consisted of three main functions of the interfacepresented as three navigation tabs at the bottom of the screen: Current Status (checking running condition), Maintenance (tracking remaining life of parts) and Planning (planning for trips based on truck condition). Since the target user is the small-scale owner, who owns a small fleet (<5 trucks), all the trucks that are part of the fleet are fixed on the screen at all times, as part of the side navigation for easy identification for the 'emergent user'.

Overview of all trucks and Instantaneous indicators
The current status tab (Figure 'Homepage'), also the homepage shows the status of all trucks - whether on a trip; waiting for trip or under maintenance along with details of the trip location; portion of trip left or maintenance activity being undertaken.
On selecting a particular truck, the values of the instantaneous indicators are represented visually along with faults in any of the other parts (if they currently exist). This is all the information the truck owner would be interested in knowing at a regular basis and hence the current status section is made the homepage. The grouping is done in this manner so that the usage of the other tabs is required only when the need arises.

Tracking health of parts
In the maintenance tab (Figure 'Health of individual part'), the owner is able to look at the health of the individual parts being represented visually in terms of 'remaining life in km' along with The 'remaining life in km' metric is used to show the health of the parts as that is how maintenance intervals are perceived in the trucking sector,



Homepage
(Summary of all trucks)



Health of individual part

ensuring the interface speaks the language of the user. Along with the health of individual parts is the service history for that particular part. This contains a list of the previously used parts of the same type and how long each of them lasted, providing the basis of a data-driven decision making process when the part is to be replaced. This can push truck owners to go for higher quality components when they begin to see long-term benefits in terms of increased life.

Visual representation of parts/instantaneous indicators
All the parts/parameters that are being monitored are represented visually (with icons/colors) throughout the interface for easy recognition and relatability to the real world (Figure 'Health of individual part'). In the case of a particular part being present at multiple places on the truck, such as brake lining which is present at all tyres of the truck, the remaining life of each of the brake linings are denoted by the fill level at their respective locations (at the respective tyres) on a frame-diagram of the chassis of the truck.

Trip planning based on truck condition
The planning tab enables the owner to decide on the next trip locations based on the condition of the truck as an added factor, making the decision making process information-driven. This is represented in terms of the maintenance activity and money that would be required (if any) for the truck with its given condition to complete a successful trip to the location that is selected on the map-based location selector (Figure 'Maintenance suggestions before trip'). This allows the owner to be informed of the maintenance that is required for the truck to successfully do a particular trip. In addition, based on the liquid cash available at the moment, the owner can choose to either carry out maintenance and do a long trip or do a shorter trip that requires lesser or no maintenance (Figures 'Maintenance suggestions before trip' & 'Running cost and risk of breakdown').

Map-based location selector
The map-based location selector (Figure 'Maintenance suggestions before trip') is a unique way to select a particular destination without the need to type a single word, developed specifically for the context of the small-scale owner who may not be familiar with typing on a smartphone. It involves an up-down sliding 'rectangle' placed on top of a map of India that has all possible trip locations marked. When the

Maintenance suggestions before trip

Running cost and risk of breakdown

'rectangle' is moved over the map, the names of the locations that fall in the area occupied by the 'rectangle' are displayed on the side of the map. From here the user can select the desired location by tapping on the desired location name.

Running cost and breakdown risk with and without maintenance
Along with showing the maintenance required to do a trip to a particular location, the interface also shows both the impact on the health of the truck (represented in terms of % risk of breakdown or 'x km' reduction in engine life) and running costs for the cases of both: maintenance being carried out and not being carried out to make the owner aware of the long-term consequences of not improper maintenance (Figure 'Running cost and risk of breakdown').

Regional language support
The interface has also been designed in one regional language – Tamil (the language of the region where user studies were conducted) as not all users were comfortable with English. With the translation, instead of going for a literal translation of all words, terms whose English versions are preferred by the target user group have been retained. For example, kilometers is represented by 'km'; "clutch plate" is spelt in Tamil instead of the actual Tamil word for it (Figure 'Homepage').

Visuals and typography
Brown, the base color on all the trucks in India, was chosen as the primary color for the interface. Two shades of brown were chosen - a light brown for fill and a bright brown for text. It was taken care that both of them match the feel of the color found on the truck. For typography, Roboto (English) and Mukta Malar (Tamil) were chosen for their readability in low font sizes and the neutral feeling they evoke.

Augsburger 101: AR tour guide app for toursits

Conference paper based on project:

Deploying Mobile Augmented Reality for Gameful Exploration of Cultural Heritage

2020. Johann Habakuk Israel/Christian Kassung / Jürgen Sieck (Hrsg.)
Kultur und Informatik: Extended Reality. vwh Verlag

*Sarah Eidenhammer, **Varun Vikash Karthikeyan**, Jayasurya Salem Sudakaran, Ankur Singh Rana, K.P. Ludwig John, Thomas Rist*

App walkthrough video
https://drive.google.com/file/d/1Evx7mcgIx1pqPdtB-RggjyWyKPOhgtI7/view?usp=sharing

Project website
http://games.hs-augsburg.de/Augsburger101/

Context

This project was done under the European Project Semester during my exchange semester in Germany. The team consisted of four international students (two students working in the design and two on the technical realization) and two professors as guides. My role was in the design part of the app and involved prototyping and refining the AR interactions along with developing the user interface for the app.

Augsburg
Augsburg is a multi-cultural town town to the south of Germany with a rich local heritage such as historical buildings, a 800-year old water management tower, signs of early industrialization and others. In addition to tourists, more than a hundred international exchange students come to attend the city's two universities every semester.

Brief
The brief was to come up with a location-based Augmented Reality game that can provide young adults a fun time walking around and exploring the city.

In addition, experience of our guide professors from an earlier project that employed NFC tags to make a treasure-hunt game suggested that longer walking paths in between 'points-of-interest' have a negative effect on the entire experience. So, it was decided that the 'points of interest' close enough and the entire experience should not last more than 30 minutes.

Process

We, the team members ourselves being part of the target demographic, started out by exploring the town ourselves and observing what we found interesting and how they could be used to make fun interactions.

Initial ideas
Post the exploration, we came up with a few initial ideas. For example, one of them was to tap into the multi-cultural aspect of the city and have virtual souvenirs at 'cultural hotspots' (places representative of each of the international cultures like vietamese restaurants, irish pubs, etc). Another one was to choose specific areas(parks, etc) and make them 'game spots' where people can go to and play casual games in a virtual-real setting(e.g, the floor is lava with AR visuals on the phone).

We felt that the idea of having a game that involved collecting virtual souvenirs to be interesting for a treasure-hunt style game. However, we also felt that it would be more relevant if the cultural hotspots/points of interest were related to the local culture of Augsburg and Germany as that is what would make the game unique to Augsburg and align better with the user's goal to explore Augsburg.

Based on our explorations of the city, we selected locations we found interesting and put them on a map. To make the process of collecting the souvenir more interesting, we drew upon from the idea of having game-like interactions in a virtual-real setting. We thought of using physical movement of the phone

in real space being the mode of interaction with the virtual space. Here again, we wanted this to be local and original to Augsburg and came up with the idea of making playful interactions out of local customs that we found to be interesting as foreigners(for example, wheat beer has its own type of glass and there is a certain way in which it is poured to avoid foam). This way, the users can have fun while also learning about local traditions. For this, we listed out local customs/traditions that we found interesting and looked for correlations with the locations on the map(for example, pretzel - a bread with very local origins and the bakery where it is made).

Out of the earlier shortlisted locations, three locations(places of interest) that were close to each other and had a correlated local custom were chosen and gameful interactions based on the correlated custom were prototyped and refined. To add coherence to these interactions, the User Interface for the app was developed(directions to places, instructions to gameful interactions, tracking progress, etc) and the prototype was tested at the locations by the team members.

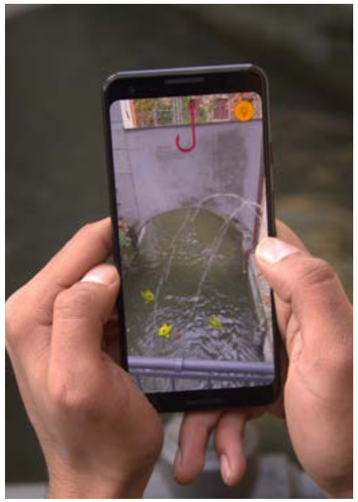
Along with the prototype, one of these gameful interactions(water tower + fishing) was made into a working model(programmed) using ARCore and Vuforia to showcase the technical feasibility.

Outcome and Key Rationales

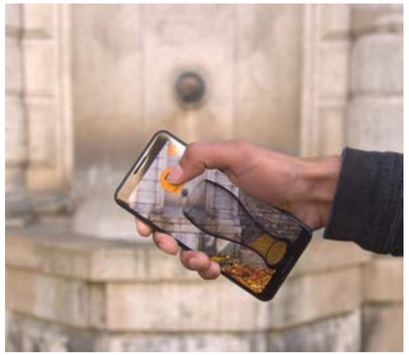
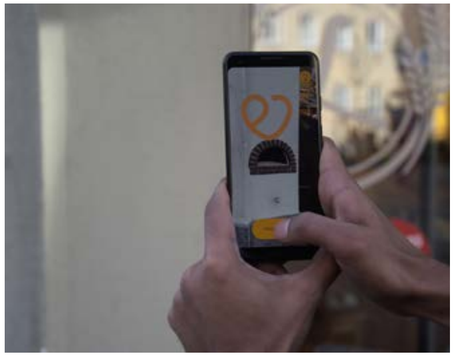
Places of Interest and gameful interactions

Water tower + Fishing
It is the part of a still-functioning 800 year-old water management system that is also a UNESCO heritage site that has a canal running in front of it. Here, the user is asked to engage in a virtual fishing exercise using a hook that is tied to the top end of the smartphone.

Historical well + Pouring wheat beer
This place is the remains of a historical well. On the app, the user can see a wheat beer dispenser in virtual space from which he/she is given a tricky beer-filling challenge to collect beer into a beer glass which is the smartphone itself.



Interaction at water tower

Interactions at historic well(L) and traditional bakery(R)

Traditional bakery + Making pretzel
The pretzel is a big part of the pastry tradition in Southern Germany. In front of the traditional bakery, the user is asked to draw a virtual pretzel by means of skillfully moving the smartphone in the air.

User Interface

Navigation compass and getting to the ‘point of interest’
A compass along with absolute distance from destination is used for navigation as this will encourage the user to look around, explore the place and find a way, making the walking process itself engaging.

The flow is non-linear, i.e., the user is free to choose which ‘point of interest’ he/she wants to go to, eliminating a dead end that may occur if the user is not able to complete one of the tasks.

Food tray progress tracker + buttons and icons
Since all of the interactions are food-related, a metaphorical plate is used as a button progress tracker. Once a particular task is completed, the respective button is replaced with a 3-dimensional virtual object that was collected from completing the gameful interaction.

Use of screen area
All the information and interactions are placed at the bottom area of the screen with a translucent background for better view of the virtual space and to facilitate one-hand use while walking.

Glow at ‘point of interest’
Upon nearing a ‘point of interest’ a virtual shining object is present at the place where the interaction is supposed to take place to let the user know where the interaction spot is. The glow vanishes once the user gets too close.

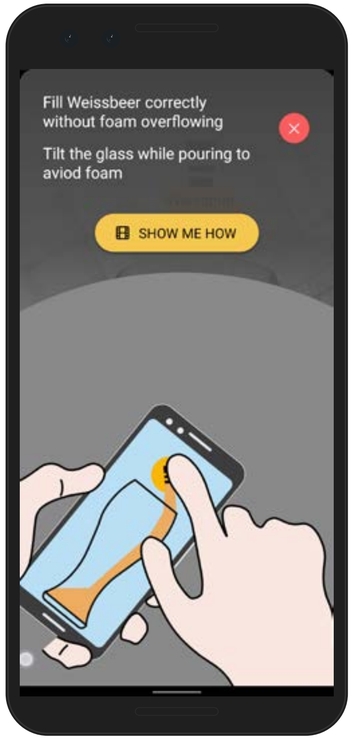
Navigation compass + Progress tracker(0/3)

Progress tracker(2/3)

Glow at ‘point of interest’

Tip button and ‘show me how’ video
A tip button is present on the screen each time the user is required to perform the gameful interaction. It gives access to the instructions on how to perform the challenge at hand. There is also a ‘show me how’ video that acts as a video tutorial for the gameful interaction that shows how the phone is to be moved.

Tip button (bulb icon)



‘Show me how’ video

ShareSpace: Service design based on shared economy

Context

This project was done as part of a course titled ‘Interactive media’ at my home university. The team consisted of 3 students. My role was in deciding on features, developing the information architecture and designing the UX part of the app.

Problem space
The team was given four problem spaces from which we had to define a problem and come up with a solution. The space we chose was to ‘Encourage critical thinking’. Within that, we chose to address the area of material consumption.

Process

Initial ideas
We started off with a few initial ideas such as a bracelet that squeezes the arm when you spend too much time on online shopping sites, an app/browser extension that shows the origin and manufacture location of the products in support of promoting local goods.

Refining brief
Soon, we realized that if we create a system in which being a critical thinker requires mental effort or self-control, then the consumer is still gullible to impulsive purchases. Hence, what we should intend to do is create a system wherein the consumer acts responsibly by just being part of the system.

Shared economy
This is where we bring in the concept of the shared economy that has gained traction in places like the USA, Canada but not so much in India where it is limited to ride sharing services and a few others. So, we wanted to scout potential and come up with a solution designed for the context.

User study
The target demographic was set to students and young professionals because of one, we thought they would be more open to the idea and two, our relative ease of access to the target group.

We spoke to 3 students and 4 early-career professionals to get their perspective and try to identify the items that they would be comfortable with sharing and the barriers that are making it difficult to share. We identified that the sense of ownership is high in India and was an important barrier for the shared economy mindset. Along with that, we also arrived at an exhaustive list of items that people would be comfortable with sharing if it was their own. In general, people were more comfortable sharing tools/equipment as opposed to something like shoes or clothing.

Co-ownership of high-quality less-use products: the ShareSpace model
To overcome the ownership barrier, we thought of a service that was based on co-ownership. This makes sense for high-quality less-use products that you wouldn’t otherwise buy owing to their low usage.

The aim is to have a digital platform where people can look for like minded people to collaboratively buy a particular product which is stored in a system of

lockers controlled by sharespace. The co-owners book their slots on the mobile app and when none of the owners are using it, it can be rented out to others to generate revenue.

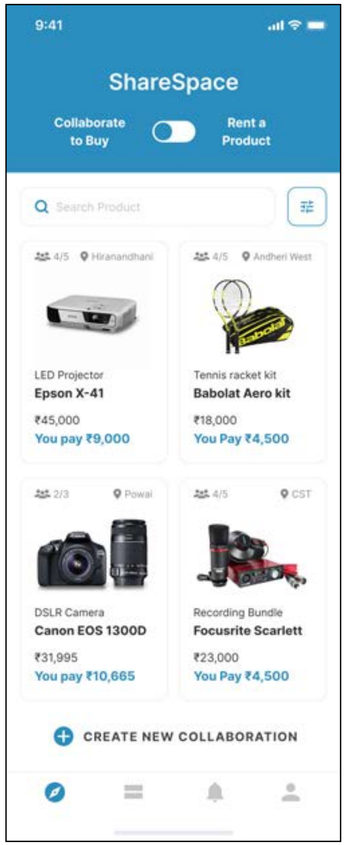
A ‘locker’ can be anything from a cupboard in an office to a dedicated warehouse that is secured by a lock with digital capabilities. The only thing is that it must be possible to individually ‘lock’ individual items. In future, the idea of ‘lockers’ can possibly be extended to spaces which can be rented on their own like say a short-term car parking spot or a ground for a game of cricket.

Prototyping the app
The interactive part of the system which is the mobile was prototyped. The features that were to be on the app were listed on individual post-its. Segregation of these features into groups led to the information architecture. Based on this, a first of wireframes was developed with medium-fidelity. These wireframes were improved and more detail was added to it, leading to the final screens.

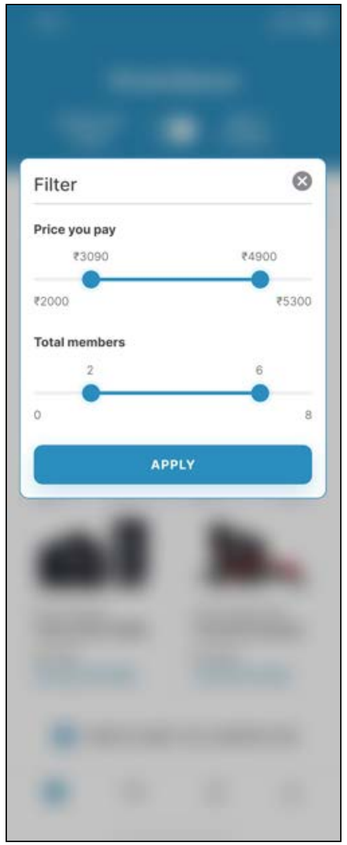
Feature sorting

Outcome and Key Rationales

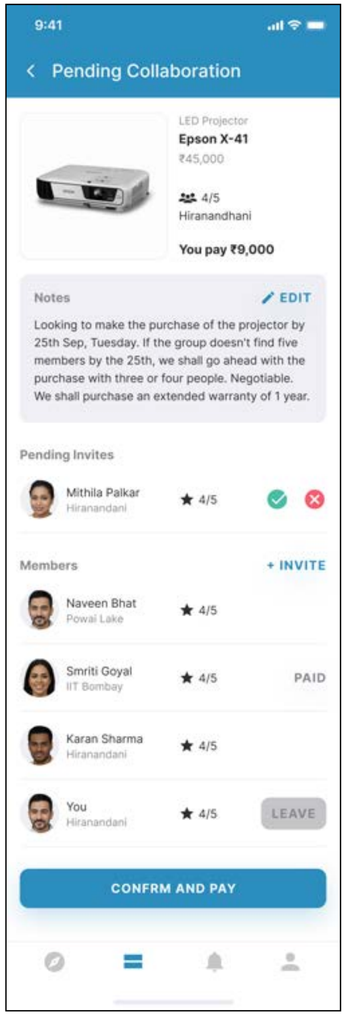
Home page
The dual nature of the service(co-buy and rent) is highlighted on the homepage by means of a toggle that switches between the collaborative buying and renting screens, giving the user an idea of what to expect.



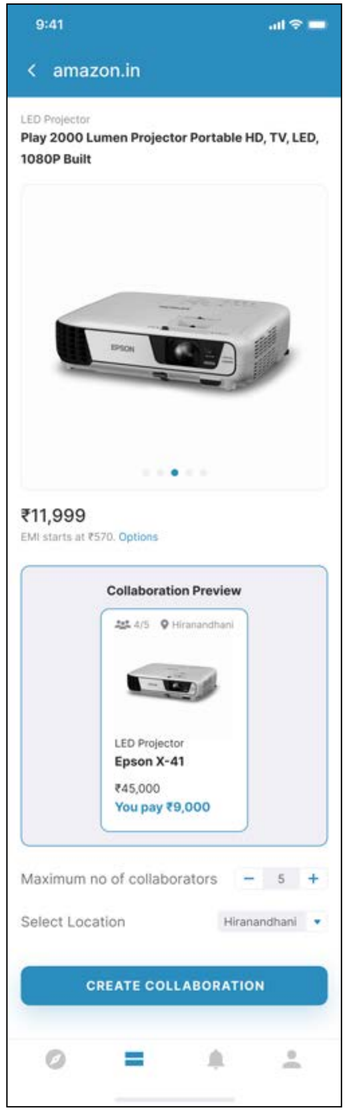
Homepage



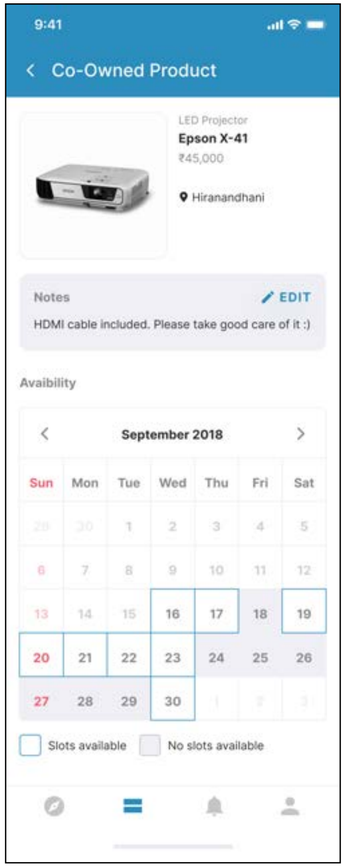
Filters to co-buy



Joining existing collaboration



Creating new collaboration



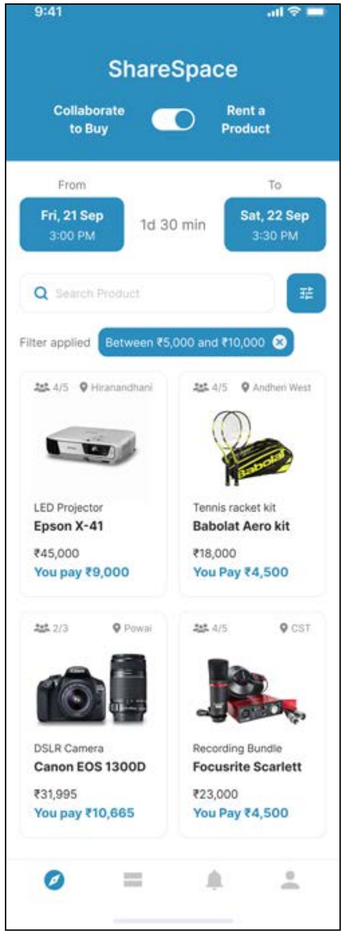
Viewing slots for co-owned product

Collaborative buying
When looking to co-buy a product, the user can look for existing groups that are looking for members to buy a particular product or create a new listing for a particular product. When the group has reached the desired number of members, they make their individual payments and the product is shipped to the 'locker' by an existing ecommerce site.

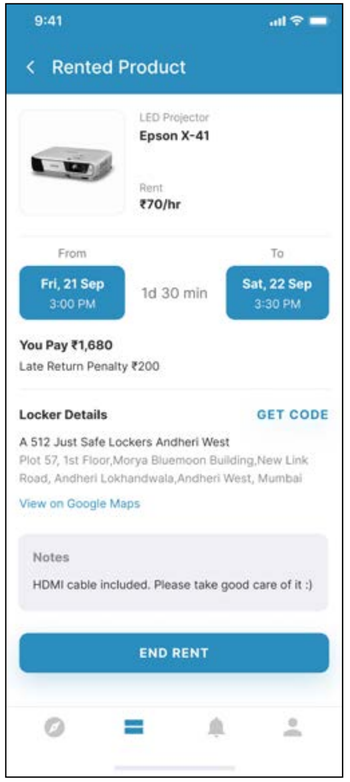
Using a co-owned product
Selecting a co-owned product shows a time-table of sorts for when that particular product is booked for use by other co-owners or people looking to rent the product. From here, the user can book slots for the use of the equipment in advance. Further the locker in which the product is stored can be accessed by entering an OTP-like changing passcode that is visible on the app once the particular user's time slot begins.

Renting a product
If the owner group wishes to make that particular product available for renting on the platform, it is made visible on the platform as 'available for rent' in the time slots that have not been booked by any of the co-owners, thereby generating revenue for them. Once rented, the access to the locker is by means of the passcode mentioned earlier. Holding on to the product beyond the rent duration is penalized with a time-based late fee to ensure punctuality.

Retaining a sense of ownership
With the idea of co-ownership, we try to retain the sense of ownership, choice of the exact model and the availability of the product. The person renting the product, can however book a slot for a product only



Products available for rent



Returning rented product

upto a week in advance so as to ensure that the owners get to book their slots first. Moreover, the owner group also has the option to make the product visible or invisible for renting. This however, shouldn't be a problem for the person looking to rent as there will be other similar products available for rent as well on the platform.

Monetary aspect

Looking at the revenue generation model of the platform, there are two aspects to it. One, a commission from the ecommerce where the product is actually shipped from and two, a percentage of the amount that is charged as rent. In future, this can be expanded to provide maintenance contracts or insurance for the products in the locker as well.

Faze: Use of face as modality for desktop control

Video of working prototype
https://www.youtube.com/watch?v=jIsuRzWoUrM

Project Report
https://drive.google.com/file/d/1h55T0wMJHR80PKqX2K8JZAIm3Z2jaqBc/view?usp=sharing

Context

This project was done during a course called Interaction Engineering during my exchange semester in Germany. It was an individual project.

I have not used images in this description as difficult to convey the idea without the video format. Kindly make use of the video links given.

Brief
The goal was to come up with a new way of interacting with the computer and also make a working prototype to demonstrate the feasibility.

Problem Space
While multitasking on a laptop, there is the possibility to make use of certain 3-finger gestures on the trackpad to quickly switch between tabs or applications. This mode of interaction is however, would not be possible if one is working on a desktop or even

While multitasking on a computer, there is a constant need to switch between applications or tabs. For example, when writing the code for a website where the user usually keeps toggling between the application in which the code is being written, a browser tab previewing the website that is being coded and a few other browser tabs to search the internet for support websites like stack overflow, w3schools and others. The three-finger gestures on the trackpad address this issue to a certain extent by enabling quick switching between applications but not between browser tabs.

However, in cases where the trackpad is not being used(user working on a desktop or simply prefer to use a mouse instead), the user will have to rely on having to move the mouse around to get the job done, which is quite a lot of movement considering the browser tabs are usually located at the top, application tabs at the bottom and the mouse is usually somewhere in the middle of the screen.

Face as modality
In such a scenario, using the face as a modality for interaction can be of value in saving time and help maintain the ‘flow’ of work. Moreover, it can also be of use to people whose physical disabilities would not allow them the luxury of working on a computer with both hands.

Process

Related work
One of the projects done in the same course during one of the previous years called FaceType, served as a starting point as to what can be done. It employed facial gestures for formatting text in a text editor (eyebrow raise = bold, purse lips = underline, tilt head = italics).

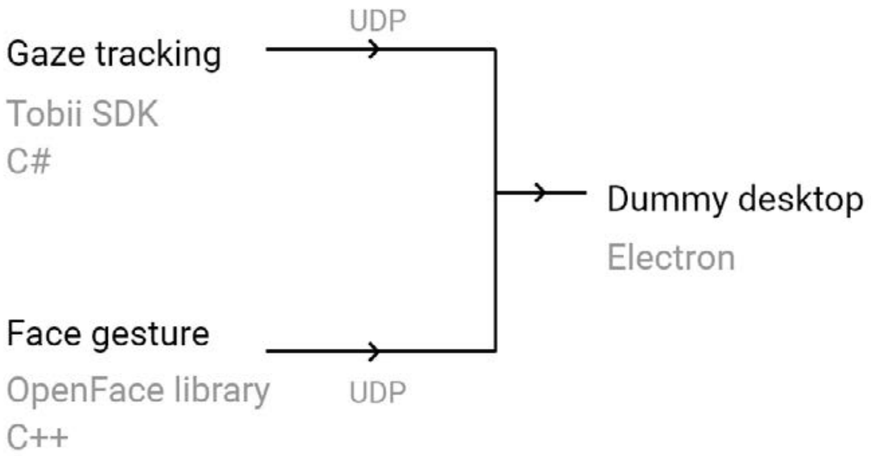
Prototyping through wizard of oz scenarios
The tasks were defined(tab switching and window switching) and various forms of interactions for these tasks were prototypes wizard of oz scenarios(involving eye movement, eyebrow movement, head tilt, lean back and forward inputs and different modes of feedback from the computer) to get a sense of things. These

interactions were corrected and refined through iterations and a final wizard of oz prototype was arrived at.

Link to the final wizard of oz prototype
https://drive.google.com/file/d/1Sz2RbOqL0p2OLVNzOU7_I-YOpEku4sV9/view?usp=sharing

Technical realization
The working prototype required three components - tracking gaze input, facial gesture recognition and a 'dummy desktop' on which the interactions can be performed.

The 'dummy desktop' was made using electron - a stripped down version of a browser with HTML and JavaScript being used to make the Graphical User Interface. Eye tracking was done using the Tobii eye tracker 4C placed at the bottom of the screen. The Tobii SDK for C# was used to transfer the gaze coordinates to the electron app via an UDP port. Facial gesture detection was done using the OpenCV's facial landmark detection library for C++ and the values were again sent to the electron app via an UDP.



Technical Realization

Outcome and Key Rationales

The interactions

1. Tab switching
Gaze input to select tab and eyebrow raise to open it.

When the user is inside an application, say, a browser, it looks like a normal browser window until the user gazes at the part of the window containing the tabs.

Once the user's gaze goes into the area containing the tabs, the tabs get extended downward to show a preview of the content of the tabs along with a visual feedback of the gaze point displayed in real-time, which stays for as long as the gaze point moves out of the area(now extended due to the tab preview) containing the tabs. Upon raising eyebrows, the tab on which the gaze point is currently on is opened.

Showing the tab preview not only gives information to the user on what is inside the tabs, but more importantly, increases the area covered by the tab on the screen, making it an easy target to hit with gaze input. Further, the visual feedback of the gaze point also lets the user know that their gaze mode is active.

While raising eyebrows is not the most natural mapping, it is something that can be learned easily and is very less energy consuming, especially when it has to be done multiple times.

2. Window switching - to adjacent window
Switching to the adjacent window is done purely with the use of tilting the head to the left or right. When the user tilts their head towards the left or right, the screen switches to the window on the corresponding side, as indicated by the order of the icons on the status bar.

Feedback to the tilt head gesture is shown in the form of a rectangular box containing all the windows appearing at the center of the screen with the selection animation moving from the current window to the window that has now been switched to by

means of the head tilt, very similar to what appears in Windows 10 on using the 3 finger left or right swipes on the trackpad.

In cases where the user wants to switch to a window that is two or more places to the left or right, then it is also possible by holding the head tilted until the selection animation moves to the window of their choice within the rectangular box containing all windows. To avoid unintended input, we can have an invisible region in the centre of the screen and have the interaction activated only when the user tilts their head while gazing at this region. There is however, no visual feedback of the gaze point that is visible here. If the user happens to tilt their head while not looking at this invisible region, the region becomes visible to the user for a few seconds as feedback, telling the user to look at this region while performing the interaction.

3. Window switching - from overview
Window switching by going to the overview involves two steps - one, leaning backwards to go to the overview and two, using gaze and eyebrow raise to select the desired window and open it, just like in the case with the tab switching.

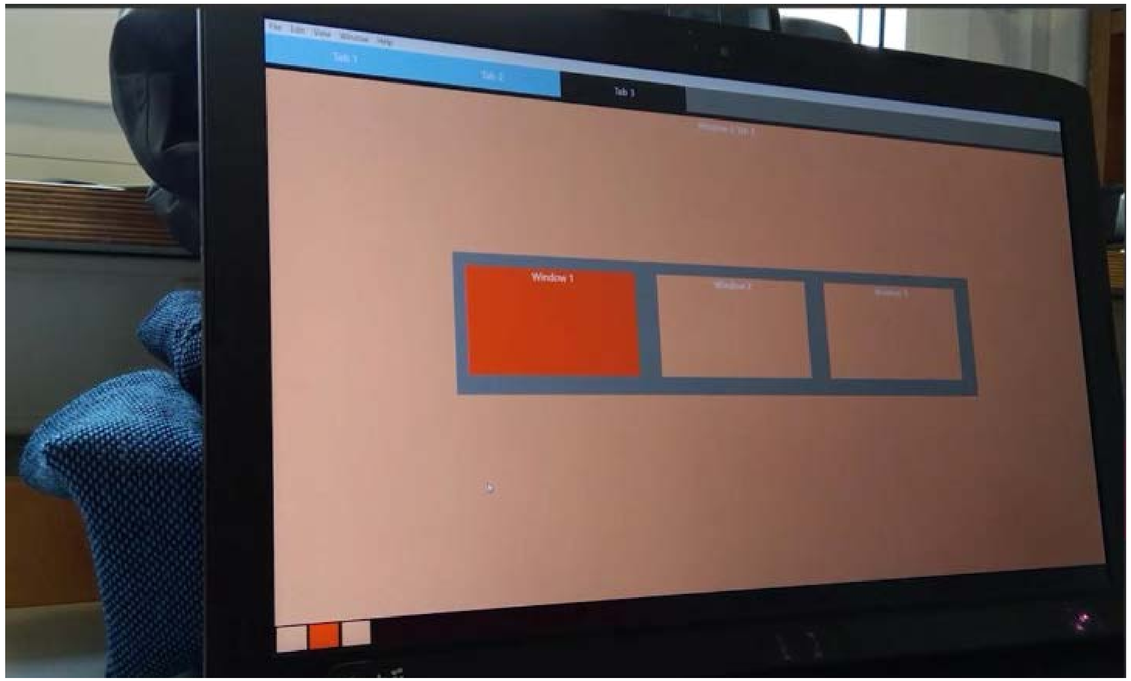
When the user leans backwards, they are taken to the overview page containing all the windows that are currently open, just like the 3-finger swipe up implementation in Windows 10. Here again, to avoid unintended input, an invisible gaze region at the centre of the screen can be implemented as in the previous case and only when the user leans back while gazing at that invisible region, the input is recorded.

Once on the overview page, visual feedback of the gaze point appears and then the user can raise their eyebrows while gazing at the desired window to open it.

Visual design of the working prototype

In the dummy desktop, all the elements were reduced to abstract elements so that there is no distraction of the other elements on the screen, which might influence the time taken by the user to perform a task.

In addition to that, all similar elements(Tabs, windows, content of the windows) were made to look similar(color, size, etc) and differentiation was made only with the help



Working Prototype

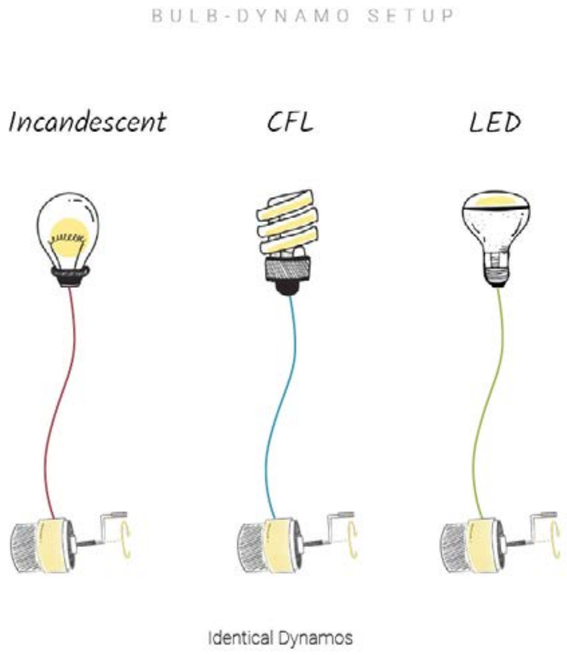
of the text on them indicating their name. This was done so that the contents of the tab/window do not have any influence.
Animations were however included in all cases as animations for hover, window shrinking into overview, switching between windows are very important to give the user feedback of what is happening and also to make the dummy desktop relatable to a real one.

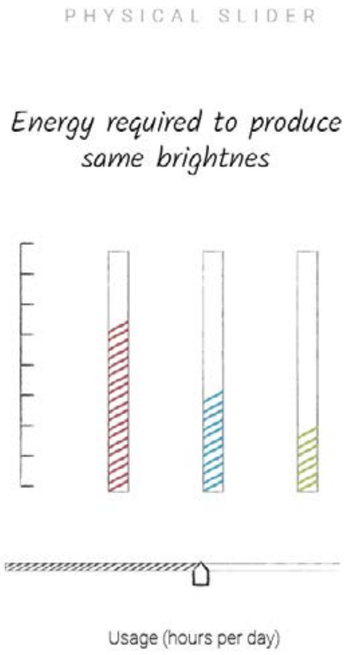
Exhibit design to show effeciency of bulbs

GIF of physical slider
https://drive.google.com/file/d/1OBsot4IQgH01oBh_EGnFUkoo0D5eq_GR/view?usp=sharing

This task was done under a course called Knowledge Organization. It was an individual task. The objective was to present a piece of information/data in a tangible form.

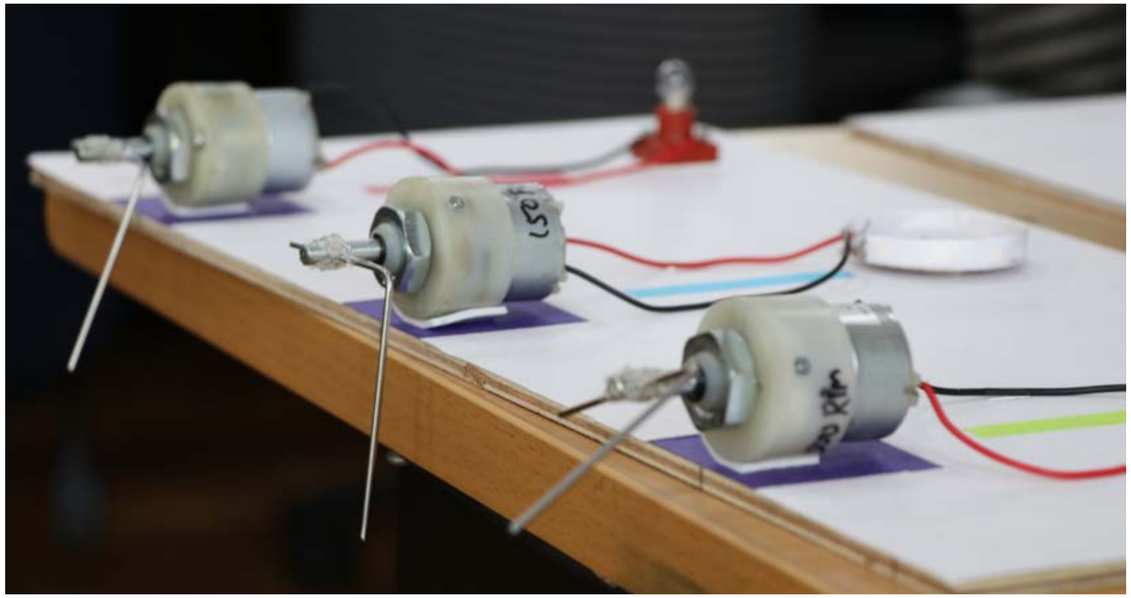
I thought of conveying information through an experience and took up the data about effeciency of different type of bulbs. The idea was to make the user power each of the bulbs using a dynamo to in order to experience the difference in effeciencies.





The setup consists of two parts - a bulb dynamo setup and a phusical interactive slider. The bulb-dynamo setup involves three identical motors connected to a different type of bulb each - LED, CFL and Incandescent giving roughly the same brightness. The motors here are used as dynamos to power the bulbs by hand. This way the user can 'feel' the effeciency of the bulbs by relating to the energy expended by him/her.

The physical slider allows the user to select the usage duration and conforms the same information with numbers to sort o round-off the experience.



Bulb-dynamo setup

Interactive Data Visualization

Link to recording of original visualization
https://drive.google.com/file/d/1UHw6L53x57zXTYwRPCatXyywuSxh6jOh/view?usp=sharing

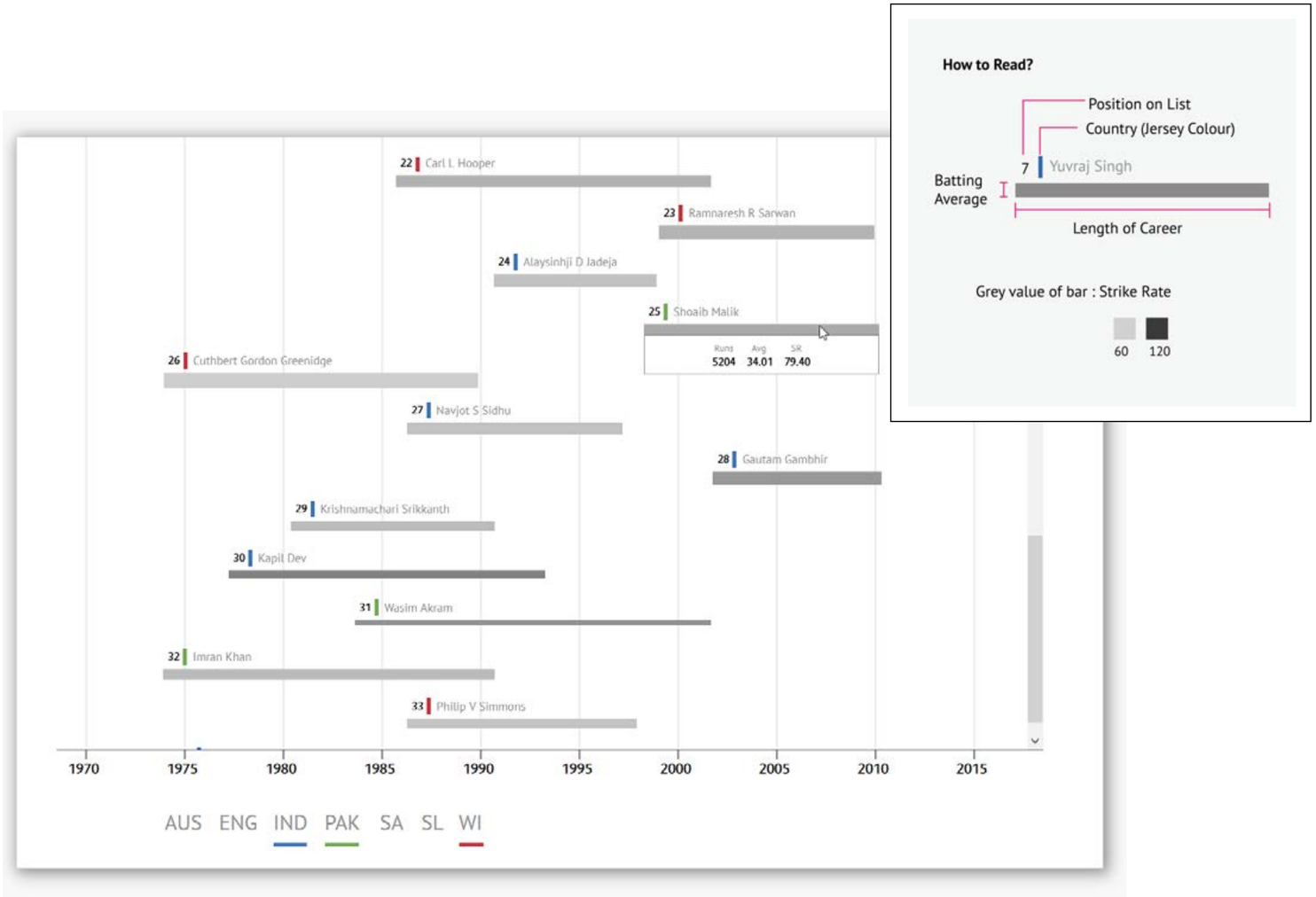
Link to recording of my redesign (this project)
https://drive.google.com/file/d/1zEU-hTsADCwdc8p_j6ckEbM8D0asUa84/view?usp=sharing

This task was done under the Data Visualization course. It was an individual task.

The visualization is the redesign of a certain 'cricket batting statistics' visualization done by bangalore-based data analytics firm Gramener.

Just for a rough introduction to what these stats mean, 'runs' is to cricket what 'points' of 'goals' are to other sports. 'Average' is the average number of runs scored before getting out and 'Strike Rate' is the rate at which these runs are scored. Average and Strike Rate are the two most common stats that are used to rate a batsman(a role like defender, midfielder, etc). In both cases, the higher the number, the better the player. The game however, has not remained the same with changes in rules and the mentality of players. Over time, it has become easier to score runs and that is reflected in the statistics as well.

However, when comparing players from different eras, it is important to keep in mind the time during which each of them played the game. This is the rationale behind the redesign and this era difference is brought into notice by placing each player's stat bars aligned to the period in which they played. Thicker the bar(y-axis), higer the Average and darker the colour, higher the Strike Rate. Jersey colours are used to denote nationalities as it is widely used by broadcasters when showing stats and will have better recall among fans.



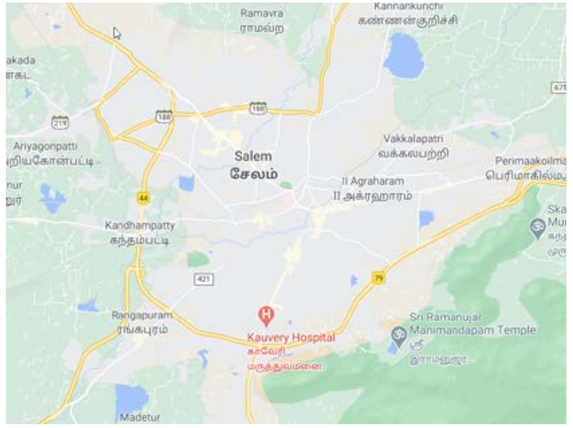
Tourist map of Salem, TN, India

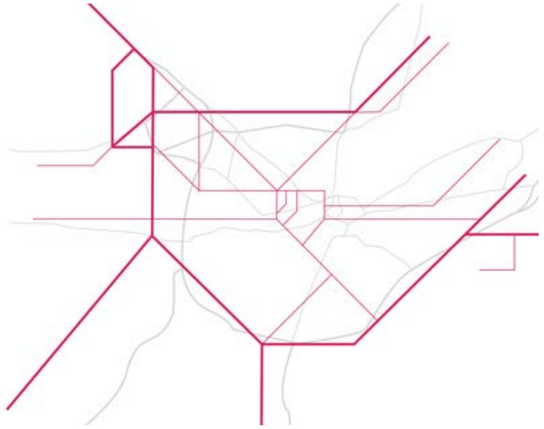
This task was done during a geo-visualization workshop at my university. It was an individual task.

I first took an image from google maps and traced the major roads. On making the lines isometric, noticed the formation of a cat-face. Though the city has nothing to do with cats, it could be helpful for tourists to make references like ‘near the ear’, etc. Hence, I made the cat-figure more prominent while staying true to geography.

Then the key locations were marked keeping the colours for different categories the same as that on google maps for easy identification. Notice the shape of the markers being consistent with the cat-shape. Also notice the text carrying a near-black shade of the icon colour for better grouping. The roads extending beyond the map are labelled to give the tourist an idea of the surrounding places.

The mininal legend in the shape of the cat-faces when placed together resemble the shape of a traveller’s souvenir that is popular in India.





Oasis: Educational board game based on nutrients and deficiencies

This project was done under the Game Design course. The team consisted of 3 members. My role was in developing the game mechanics and the designing the game board and player card-holders.

The aim of the game is to educate middle-school children about nutrients and deficiency diseases. The game was developed and refined over multiple play testing sessions with kids. It is designed for 3-6 players.

It is a hunter-gatherer style strategy game in which players have to move around the board collecting food items and keep themselves nourished on four nutrients - Vitamins A, B, C and Iron.

If a player is short of a particular nutrient, then the deficiency disease corresponding to that particular nutrient comes into play. It was ensured that the effects of deficiency diseases in the game were related to each of their effects in real life. For example, deficiency of Vitamin A(night blindness) means the player can only move half the number of steps. Deficiency of Iron(anaemia) means the player cannot attack the other players and so on.

The effects of these deficiencies can however, be overcome by collecting food with that particular nutrient, just like in real life. The packaging for the game was also made complete with a rule book.